

Exploratory Data Analysis (EDA) - Detailed Interview Questions & Answers

1. What is EDA and why is it important?

Exploratory Data Analysis (EDA) is the process of visually and statistically examining datasets to discover patterns, spot anomalies, test assumptions, and check relationships between variables. It is important because it allows analysts to:

- Understand the structure and distribution of data.
- Detect missing values and outliers.
- Form hypotheses and guide feature engineering.
- Provide business insights before applying advanced models.

2. Which plots do you use to check correlation?

To check correlation between variables:

- Heatmap: Shows correlation coefficients in a matrix form (e.g., Pearson correlation).
- Pairplot: Visualizes pairwise scatterplots and distributions.
- Scatterplot: Useful for examining correlation and linear/non-linear trends between two variables.

3. How do you handle skewed data?

Skewed data can distort statistical analysis and model performance. Ways to handle it:

- Apply mathematical transformations: log, square root, Box-Cox, Yeo-Johnson.
- Use robust statistics like median and IQR instead of mean and standard deviation.
- Convert continuous skewed variables into categorical bins.
- For machine learning: prefer algorithms like Random Forest or XGBoost which are less sensitive to skewness.

4. How to detect multicollinearity?

Multicollinearity occurs when two or more predictor variables are highly correlated, leading to instability in regression models. Detection methods:

- Correlation Matrix: Look for correlation values above 0.8 or 0.9.
- Variance Inflation Factor (VIF): $VIF > 10$ usually indicates multicollinearity.
- Eigenvalues/Condition Index: High values indicate dependencies among features.

5. What are univariate, bivariate, and multivariate analyses?

- Univariate Analysis: Analysis of a single variable, e.g., histogram, bar chart, boxplot.
- Bivariate Analysis: Analysis of two variables, e.g., scatterplot, correlation, grouped boxplots.
- Multivariate Analysis: Analysis involving more than two variables, e.g., pairplot, PCA, regression models, clustering.

6. Difference between heatmap and pairplot?

- Heatmap:
 - * Shows correlation coefficients in a grid format.
 - * Best for summarizing relationships in large datasets.
- Pairplot:

- * Displays scatterplots between every pair of variables and histograms on the diagonal.
- * Useful for detecting trends, clusters, and distributions.

7. How do you summarize your insights?

When summarizing insights from EDA:

- Highlight key patterns (e.g., in Titanic dataset: women and children had higher survival rates).
- Mention anomalies or outliers (e.g., unusually high fares).
- Support statements with visuals and statistical evidence.
- Provide actionable insights or hypotheses for modeling (e.g., passenger class and gender are strong predictors of survival).